TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

–1 POINTS FOR EACH FIELD LEFT EMPTY OR FILLED WRONG

FIRST NAME: LAST NAME:

ID NUMBER: SECTION:

STUDENT SIGNATURE: ROOM NUMBER

MIDTERM EXAMINATION

COURSE NAME: DISCRETE MATHEMATICS

COURSE NUMBER: MSC 2050

EXAMINATION DURATION: 1 HOUR 30 MINUTES

ADDITIONAL MATERIALS ALLOWED TO BE USED:

SPECIAL INSTRUCTIONS

(1) FOLLOW ANY INSTRUCTIONS GIVEN BY PROCTORS. COMPLIANCE IS REQUIRED! DO NOT ARGUE WITH A PROCTOR, YOU GET "F" FOR MISBEHAVING + other possible complications.

(2) ONCE YOU ARE ALLOWED TO OPEN THE EXAMINATION BOOKLET, YOU MUST PUT ONLY YOUR **ID** ON EVERY ODD PAGE (do NOT write your name)

- Final answers must be written by **only blue or black, non-erasable pen**.
- Do NOT use highlighters or correction pen.
- IUT's Examination Rules and Regulations apply.

Do NOT open the examination paper until directed to do so!

1. (4 points)Find the DNF of $((p \oplus q) \rightarrow (r \wedge q)) \vee (p \wedge r)$ by following the steps:

   a) (2 points)Construct the truth table of this compound proposition.

   b) (2 points)Write the DNF according to the table.

2. (6 points)Encircle the true statements. There are penalty points for wrong answers.

   - Cardinality of the union of two finite sets is equal to the sum of their cardinalities
   - Countable union of countable sets is also countable.
   - Power set of countable set is also countable.
   - If $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$ then there is onto function from B to A.
   - Any subset of uncountable set is also uncountable.
   - If $A = \{a, b, c\}$ and $B = \{a, b\}$ then $A \cap B = B$
   - The set of rational numbers is uncountable set.
   - There is a one-to-one correspondence from $\mathbb{Q}$ to $\mathbb{N}$

3. (7 points)Solve the problems given below:

   a) (4 points)Write an algorithm in pseudocode which multiplies two 3×3 matrices.

   b) (3 points)Give big-O estimate for the number of additions.

4. (8 points)Find $11^{644} \mod 645$ using the algorithm of Modular exponentiation by following the steps:
   a) (1 point)Find binary expansion of 644
   b) (2 points)How many steps in this algorithm(loop)?
   b) (3 points)Write each step of the algorithm into table.
   c) (2 points)Can we apply Fermat's little theorem to this problem? Justify.

$644 = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\__2$

$x = \_\_\_\_ power = \_\_\_\_\_\_\_\_\_\_\_\_$

i	a_i	x	$power$

5. (5 points)Encrypt the message STOP POLLUTION using Caesar cipher for k=23:

A	B	C	D	E	F	G	H	I	J	K	L	M
$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$
13	14	15	16	17	18	19	20	21	22	23	24	25

Do NOT fill	P.1	P.2	P.3	P.4	P.5	Total
max points	4	6	7	8	5	30
your score						